

Gate E5 report — run_v1.22_r1 (Paper IV v1.22-r1)

Verdict: **PASS**. Evidence: 20_EVIDENCE/E5/. Auditor: Claude Opus 5.5, same session as run_v1.2_r1/run_v1.21_r1; no Lean executed.

E5 — Exact arithmetic (no Lean)

Script	Status this cycle	Result
e5_v122_c3.py → E5_V122_C3.json (new, auditor-written)	Executed	14 checks (C3a–C3n) of the new C.3 text: all PASS. 8 of them carry an inline negative control. 7 of those controls were rejected; the C3m control was mis-specified, because its variant 2^{29R} is also true (CORRECTIONS C2-01).
e5_v122_c3_negfix.py → E5_V122_C3_negfix.json	Executed	Corrected C3m control: $(R \cdot 16^R)^6 \leq 2^{24R}$ is false on 1..50 and is rejected as required.
e5_v122_c3_neg_supplement.py → E5_V122_C3_neg_supplement.json	Executed	Negative controls for the six checks that had none (C3c, C3e, C3g, C3j, C3l, C3n): all 6 perturbed claims are false and rejected. With this file, every C.3 check has a rejected negative control.
e5_v121_numerics.py → E5_V121_RERUN.json (copied from run_v1.21_r1)	Repeated	The 21 v1.21 checks (C.2, the C.3 schedule and selector chain, E.1, E.2d): all PASS, all negatives rejected. Table C.1 of C.3 is unchanged.
run_v1.2_r1 T01–T24	Inherited	The covered text is unchanged (diff).

Typing of the evidence

- C3a–C3g are exact rational and integer identities and inequalities at the fixed parameters, computed with `fractions.Fraction` and Python big integers.
- C3h, C3i, C3l, C3m and C3n check elementary universal lemmas only on finite ranges: $x \cdot 4^x \geq 2^x$, $16^x \geq 4x$, $8q \leq 2^q$ ($q \geq 6$), $256B^2 \leq B^6$ ($B \geq 4$), $4x \cdot 4^x \leq 2^{\{4x\}}$, $R^6 \leq 2^{\{6R\}}$, and $30R \leq 8q$ given $4R \leq q$.
This is a finite regression, not a universal proof.
 - The universal statements were proved by hand in E2; they are one-line inductions or monotonicity arguments.
 - They are also covered by the frozen E19 theorems, whose build E4 confirms by reuse.
- Quantities such as R, B and $T(h+7)$ are never materialised; the argument works on exponents.

E5 verdict: PASS.